

Study Questions: Agents (Fiber & Flow)

1. In what sense are a crocheter's hands "agents" rather than passive tools? What properties of agency do they exhibit?
2. How does the density of mechanoreceptors in the fingertips contribute to the hands' ability to function as active inference agents when working with yarn?
3. Compare a beginner crocheter's hands to an expert's hands in terms of their generative models. What does the expert's model contain that the beginner's does not?
4. Explain precision weighting using the example of an experienced crocheter picking up an unfamiliar fiber for the first time. What changes in how the hands process sensory information?
5. Why do experienced crafters say they can "feel" a knot in the yarn before they see it? What Active Inference mechanism explains this ability?
6. Describe the perception-action loop as it operates in a single crochet stitch. What does the hand sense, predict, and do?
7. What does it mean to say that crocheting is a "dialogue" between agent and material? Give a specific example of how yarn "pushes back" against the crocheter's actions.
8. How does a slippery silk yarn change the precision weighting of a crocheter who usually works with cotton? What adjustments must the agent make?
9. Explain the nested hierarchy of agents in a crocheter (fingertip, hand, arm, whole person). How do these levels coordinate?
10. Why do crafters describe certain yarns as having "personality"? Translate this folk observation into Active Inference language.
11. How is the development of tactile expertise in crochet similar to the development of expertise in other skilled manual practices (pottery, surgery, cooking)?

12. What role does the non-dominant hand play as an agent in crochet? How does its function differ from the hook hand?
13. A crocheter switches from a 5mm hook to a 3.5mm hook with the same yarn. How does this tool change affect the hand-agents' generative models and predictions?
14. Describe what happens — in terms of prediction error and free energy — when a crocheter encounters a sudden change in yarn thickness (a "slub") while working.
15. How might arthritis or hand fatigue change the precision weighting of the hand-agents? What compensatory strategies might the crocheter adopt?
16. Why do many crocheters prefer to shop for yarn in person rather than online? Frame your answer in terms of the embodied agent's need for direct sensory evidence.
17. How does muscle memory in crochet relate to the concept of a generative model? Is muscle memory a form of prediction?
18. Describe a situation where the crocheter's hands and eyes disagree about the state of the work. Which agent typically detects errors first, and why?
19. How might the concept of "active sensing" apply to the way a crocheter rolls yarn between their fingers to assess its quality?
20. If you were teaching a beginner to crochet, how would you help them begin building a generative model of yarn behavior in their hands? What experiences would be most valuable?